Kinetic lattice grand canonical Monte Carlo simulation for ion current calculations in a model ion channel system

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An algorithm in which kinetic lattice grand canonical Monte Carlo simulations are combined with mean field theory (KLGCMC/MF) is presented to calculate ion currents in a model ion channel system. In this simulation, the relevant region of the system is treated by KLGCMC simulations, while the rest of the system is described by modified Poisson-Boltzmann mean field theory. Calculation of reaction field due to induced charges on the channel/water and membrane/water boundaries is carried out using a basis-set expansion method [Im and Roux, J. Chem. Phys. **115**, 4850 (2001)]. Calculation of ion currents, electrostatic potentials, and ion concentrations, as obtained from the KLGCMC/MF simulations, shows good agreement with Poisson-Nernst-Planck (PNP) theory predictions when the channel and membrane have the same dielectric constant as water. If the channel and membrane have a lower dielectric constant than water, however, there is a considerable difference between the KLGCMC/MF and PNP predictions. This difference is attributed to the reaction field, which is missing in PNP theory. It is demonstrated that the reaction field as well as fixed charges in the channel play key roles in selective ion transport. Limitations and further development of the current KLGCMC/MF approach are also discussed. © *2007 American Institute of Physics*. [DOI: 10.1063/1.2748373]

I. INTRODUCTION

Ion transport through ion channels in cell membranes is of crucial importance in biology and biophysics as a result of its role in electrical signaling in the nervous system and in a variety of metabolic functions.[1,2] Because of this importance, many theoretical and computational approaches as well as experimental studies have been devoted to the understanding of ion transport.[3–17] Among the theoretical and computational approaches is Poisson-Nernst-Planck (PNP) mean field theory, which is based on a dielectric continuum model.[7,11,17] In PNP theory, proteins, membranes, ions, and water are considered as continuum dielectric materials, and the atomistic details of the proteins and membranes are ignored. Although the method has proven to describe ion transport reasonably well with appropriate dielectric constants and ion diffusion coefficients, an important limitation is the absence of a reaction field effect, which is something that could be important when ions enter a low dielectric medium such as an ion channel or membrane.[14,18–20] Also ignored in the PNP theory are the correlation effects of ions. On the other hand, molecular dynamics (MD) simulations can be used to elucidate ion transport through ion channels at an atomistic level with an assumed force field.[21–23] However, the number of atoms and time scales involved in ion transport still make full atomistic MD simulations computationally demanding.[24]

As alternatives, kinetic (or dynamic) Monte Carlo (KMC or DMC) simulations and Brownian dynamics (BD) simulations can bridge the gap between PNP theory and atomistic MD simulation.[6,8,10,14,15,25,26] KMC and BD simulations can include reaction field effects by using basis-set expansion methods[9] or other methods.[8,16,27,28] In addition, ion correlation effects are included in KMC and BD simulations by employing explicit ions. Although KMC and BD simulation methods are also based on a dielectric continuum description of the proteins, membranes, and water, this enables the KMC and BD simulations to reach time scales that are much longer than can be studied using MD simulations.

Because of their advantages over PNP and MD methods in the study of ion channel transport, KMC and BD simulations have been very popular. Moy *et al.* used BD simulations to test the validity of Poisson-Boltzmann (PB) or PNP theory by comparing predictions with BD results.[6,25] A dynamic lattice Monte Carlo simulation approach was presented by Graf *et al.* to describe ion transport in a model ion channel system.[8,14] Im *et al.* developed a method which combines BD simulations with grand canonical Monte Carlo calculations (BD/GCMC algorithm).[10,26] They used that method to evaluate ion currents through the OmpF Porin channel.[10] Finally, Cheng *et al.* performed DMC simulations for the calculation of ion currents and prediction of the structure of an ion channel.[15]

In this paper, we present a hybrid approach, denoted as KLGCMC/MF, in which the relevant region of the system including ion channels is treated with kinetic lattice grand canonical Monte Carlo (KLGCMC) simulations and the rest of the system is described by a modified PB mean field (MF)

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theory. Based on a lattice algorithm, the KLGCMC/MF simulation method reduces computational time by avoiding any interpolation scheme that is otherwise needed to calculate electrostatic potentials or forces in off-lattice simulations. To realize steady-state dynamics where ions flow at a constant rate, ions are created and deleted at restricted regions using a grand canonical MC scheme with a local control method.[29–32] The reaction field effect in the KLGCMC/MF simulation method is treated by a basis-set expansion method which was proposed by Im and Roux.[9] We test the KLGMC/MF simulation method by calculating ion currents for K^+ and Cl^- in a model ion channel system and comparing simulation predictions with PNP calculations. Comparison of the KLGCMC/MF simulation results with the PNP calculations is also employed to investigate the effect of the reaction field on ion currents.

This paper is organized as follows. In the next section, a KLGCMC/MF algorithm is proposed. The basis-set expansion method for the reaction field is also reviewed briefly. In Sec. III, a model ion channel system is introduced and the KLGCMC/MF simulation method is used to calculate ion currents and other properties in the model system. Comparison of KLGCMC/MF simulation results with PNP calculations is also presented in the same section. Limitations and further studies of the current KLGCMC/MF approach are discussed in the final section.

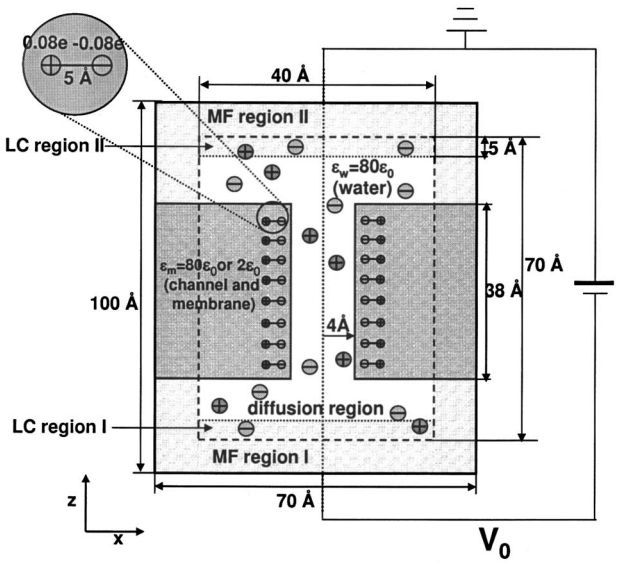


FIG. 1. Schematic of a model ion channel. The whole system is composed of a lattice grand canonical Monte Carlo (LGCMC) region and two mean field (MF) regions. In the MF region I and II, the ions are described by the screening factor $\kappa_{I(II)}(\mathbf{r})$ and the explicit ions appear only in the LGCMC region. The LGCMC region comprises two local control (LC) regions and one diffusion region. LC regions I and II in the LGCMC region are located at the bottom (-35 Å$\leqslant z\leqslant -30$ Å) and the top (30 Å$\leqslant z\leqslant$35 Å), respectively.

II. METHODS

A. KLGCMC/MF simulation with the local control method

In the present KLGCMC/MF simulation, the whole system is divided into three regions: two MF regions and a lattice grand canonical Monte Carlo (LGCMC) region (Fig. 1). In the MF regions, electrostatic potentials and ion concentrations are obtained by solving modified PB equations as will be explained in Sec. II B. In the LGCMC region, cations and anions are created or deleted at a lattice point, or moved from one lattice point to another at each time step. In the original GCMC simulation method by Valleau and co-workers, cations and anions were created or deleted globally.[29–31] This method, referred to as the global GCMC method, however, causes difficulties in calculating ion currents in a small region such as an ion channel. The creation and deletion of ions in highly confined regions leads to large number fluctuations, which result in large statistical errors in the calculation of ion currents. To avoid large statistical errors in the ion current calculation, the KLGCMC/MF simulation in this study employs a local control method proposed by Papadopoulou *et al.*[32] In the local control method, the creation and deletion of ions are restricted to local control (LC) regions (or buffer regions[9]) far from the region of interest, and only diffusion of ions occurs in the diffusion region. For the model system in this work, the LC regions are located at the bottom and top of the LGCMC region shown in Fig. 1.

In the KLGCMC/MF simulation with the local control method, ion creation is accomplished by randomly selecting a lattice site in one of the two LC regions and attempting to create a cation or anion at that lattice point. A creation attempt is accepted or rejected with a probability of $P_{N_L\rightarrow N_L+1}$,

LC region I:

$$P_{N_L\rightarrow N_L+1} = \min\left\{1, \frac{\rho_v}{N_L+1}\exp[B-\beta(U_{N_L+1}-U_{N_L}-q_{\mathrm{cre}}V_0)]\right\}, \tag{1a}$$

LC region II:

$$P_{N_L\rightarrow N_L+1} = \min\left\{1, \frac{\rho_v}{N_L+1}\exp[B-\beta(U_{N_L+1}-U_{N_L})]\right\}. \tag{1b}$$

Here $\rho_v \equiv V_L/V$ where V_L and V are the volumes of that LC region and the simulation box, respectively. N_L (or N_L+1) is the ion number in that LC region before (or after) a creation attempt, U_N (or U_{N+1}) is the total interaction energy of ions before (or after) the creation attempt, q_{cre} is the charge of the created ion, and V_0 is the applied potential shown in Fig. 1, where a voltage of V_0 is applied to the bottom and a zero voltage is applied to the top of the system. β is $1/k_BT$ where k_B is the Boltzmann constant and T is the temperature.

The constant B in Eqs. (1a) and (1b), which is associated with the bulk ion concentration, is given by

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TABLE I. Ion parameters.

Ion	$\epsilon^{\mathrm{LJ}a,b}$ (kcal/mol)	$\sigma^{\mathrm{LJ}a,b}/2$ (Å)	D^{c} ($\times 10^{-5}$ cm^2/s)	B [Eq. (2)] 0.10M KCl	B [Eq. (2)] 0.50M KCl
K^+	0.087	1.5713	2.0	2.37	3.79
Cl^-	0.150	2.0220	2.0	2.37	3.79

[a]Taken from Ref. 37.
[b]$\epsilon_{ij}^{\mathrm{LJ}}=\sqrt{\epsilon_{ii}^{\mathrm{LJ}}\epsilon_{jj}^{\mathrm{LJ}}}$ and $\sigma_{ij}^{\mathrm{LJ}}=(\sigma_{ii}^{\mathrm{LJ}}+\sigma_{jj}^{\mathrm{LJ}})/2$.
[c]Taken from Ref. 17.

$$B \equiv \beta\mu + \ln\frac{V}{\Lambda^3}, \tag{2}$$

where μ is the chemical potential of the created ion and $\Lambda \equiv h/(2\pi m k_B T)^{1/2}$ where m is the mass of the created ion. Here we assume that cations and anions have the same chemical potential and mass.

For ion deletion, an ion in one of the two LC regions is randomly selected and an attempt to delete that ion is accepted or rejected by generating and comparing a random number with a probability of $P_{N_L\to N_L-1}$,

LC region I:

$$P_{N_L\to N_L-1} = \min\left\{1, \frac{N_L}{\rho_v}\exp[-B-\beta(U_{N_L-1}-U_{N_L}+q_{\mathrm{del}}V_0)]\right\}, \tag{3a}$$

LC region II:

$$P_{N_L\to N_L-1} = \min\left\{1, \frac{N_L}{\rho_v}\exp[-B-\beta(U_{N_L-1}-U_{N_L})]\right\}, \tag{3b}$$

where q_{del} is the charge of the deleted ion.

An ion move is performed by sequentially selecting an ion in the LGCMC region and moving that ion to one of the six nearest-neighbor sites in the case of a three-dimensional (3D) simulation. The move is accepted or rejected with a probability $P_{a\to b}$ given by

$$P_{a\to b} = \min\{1, \exp[-\beta(U_{N_b}-U_{N_a})]\}, \tag{4}$$

where U_{N_a} (or U_{N_b}) is the total interaction energy of N ions before (or after) a move and a (or b) is the ion position before (or after) the move.

B. Calculation of the ion interaction energy

In this model system, ions interact with the fixed charges in the ion channel as well as with other ions. Ions also interact with an applied external field. The interaction of ions with the reaction field due to induced charges at the channel/water and membrane/water boundaries is also taken into account. As a result, the total interaction energy U_N in the system is given as[9]

$$U_N(\mathbf{r}_1,\mathbf{r}_2,\ldots,\mathbf{r}_N) = \sum_{i<j} u_{ij}(\mathbf{r}_i,\mathbf{r}_j) + \sum_i h_i(\mathbf{r}_i) + \sum_i q_i\phi_{\mathrm{sf}}(\mathbf{r}_i) + \frac{1}{2}\sum_i q_i\phi_{\mathrm{rf}}(\mathbf{r}_i). \tag{5}$$

A detailed explanation of each term on the right-hand side of Eq. (5) is given elsewhere,[9,27] so we only give a brief description here. In Eq. (5), the direct interaction between ions i and j, $u_{ij}(\mathbf{r}_i,\mathbf{r}_j)$, is given by

$$u_{ij}(\mathbf{r}_i,\mathbf{r}_j) = \frac{1}{4\pi\epsilon_w}\frac{q_iq_j}{r_{ij}} + 4\epsilon_{ij}^{\mathrm{LJ}}\left[\left(\frac{\sigma_{ij}^{\mathrm{LJ}}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}^{\mathrm{LJ}}}{r_{ij}}\right)^{6}\right], \tag{6}$$

where $r_{ij}=|\mathbf{r}_i-\mathbf{r}_j|$, ϵ_w is the water dielectric constant in bulk, q_i and q_j are the charges of ions i and j, and $\epsilon_{ij}^{\mathrm{LJ}}$ and $\sigma_{ij}^{\mathrm{LJ}}$ are parameters associated with the Lennard-Jones (LJ) potential (See Table I). The interaction energy $h_i(\mathbf{r}_i)$ in Eq. (5) represents the hard-wall interaction of ion i with the channel/water and membrane/water boundaries and is given by

$$h_i(\mathbf{r}_i) = \begin{cases} \infty & \mathbf{r}_i \in \text{channel or membrane region} \\ 0 & \text{otherwise.} \end{cases}$$

This term also includes the hard-wall interaction of ion i with the boundaries between the LGCMC region and the MF regions. The electrostatic potential $\phi_{\mathrm{sf}}(\mathbf{r}_i)$, which accounts for the interaction between ion i and the fixed charges in the channel and membrane, is calculated with the modified PB equations.[9,26,33,34] The modified PB equations for the LGCMC region, MF region I, and MF region II, respectively, read

LGCMC region:

$$\nabla\cdot(\epsilon(\mathbf{r})\nabla\phi_{\mathrm{sf}}(\mathbf{r})) = -\rho_f(\mathbf{r}), \tag{7a}$$

MF region I:

$$\nabla\cdot(\epsilon(\mathbf{r})\nabla\phi_{\mathrm{sf}}(\mathbf{r})) - \epsilon_w\kappa_{\mathrm{I}}^2(\mathbf{r})(\phi_{\mathrm{sf}}(\mathbf{r})-V_0) = -\rho_f(\mathbf{r}). \tag{7b}$$

MF region II:

$$\nabla\cdot(\epsilon(\mathbf{r})\nabla\phi_{\mathrm{sf}}(\mathbf{r})) - \epsilon_w\kappa_{\mathrm{II}}^2(\mathbf{r})\phi_{\mathrm{sf}}(\mathbf{r}) = -\rho_f(\mathbf{r}), \tag{7c}$$

Here $\epsilon(\mathbf{r})$ and $\kappa_{\mathrm{I(II)}}^2(\mathbf{r})$ are the space-dependent dielectric constant and Debye-Hückel screening factor, respectively. The Debye-Hückel screening factor is defined as $\kappa_{\mathrm{I(II)}}^2(\mathbf{r}) \equiv 2\beta|e|^2c_{\mathrm{I(II)}}^0H(\mathbf{r})/\epsilon_w$ for a monovalent ionic solution where $c_{\mathrm{I(II)}}^0$ is the bulk ion concentration in the MF region I(or II)

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and $H(\mathbf{r})=1$ is for ion-accessible regions and otherwise $H(\mathbf{r})=0$. In Eqs. (7a)–(7c), $\rho_f(\mathbf{r})$ represents the charge density for the fixed charges in the ion channel and membrane. In Eq. (7c), V_0 is an external potential applied to the system.

The last term in Eq. (5) deals with the interaction of ion i with the induced charges on the channel/water and membrane/water boundaries. The charges are induced when an ion enters the channel or approaches close to the membrane/water boundary. Because ion channels and membranes generally have lower dielectric constants than water, the induced charges have the same sign as the entering or approaching ion charge. As a result, the repulsive interaction between the ion and the induced charges decreases the ion currents and ion concentrations inside the channel. This so-called reaction field effect or dielectric self-energy effect is included in all of our simulations.[9,14,19,20,28]

To calculate the reaction field potential $\phi_{\mathrm{rf}}(\mathbf{r})$, we employ the basis-set expansion method proposed by Im and Roux.[9] In this method, $\phi_{\mathrm{rf}}(\mathbf{r})$ is expressed as a linear combination of the basis-set electrostatic potentials $\phi_n^{\mathrm{rf}}(\mathbf{r})$,

$$\phi_{\mathrm{rf}}(\mathbf{r}) = \sum_n^N C_n \phi_n^{\mathrm{rf}}(\mathbf{r}). \tag{8}$$

Here N is the number of basis functions and the coefficient vector $\mathbf{C}$ is given by

$$\mathbf{C} = \mathbf{S}^{-1}\mathbf{Q}, \tag{9}$$

where the elements of the $N \times N$ overlap matrix $\mathbf{S}$ and the generalized multipole moment vector $\mathbf{Q}$ are written as

$$S_{mn} = \int d\mathbf{r} b_m(\mathbf{r}) b_n(\mathbf{r}), \tag{10}$$

$$Q_n = \sum_i q_i b_n(\mathbf{r}_i), \tag{11}$$

respectively. In the present KLGCMC/MF simulation for an ion channel, we use the Legendre polynomials as the basis functions, which are given by[35]

$$b_n(\mathbf{r}) = H(\mathbf{r}) \left[\frac{(2p+1)}{L_x} \frac{(2q+1)}{L_y} \frac{(2r+1)}{L_z} \right]^{1/2} \times P_p\left(\frac{2}{L_x}x\right) P_q\left(\frac{2}{L_y}y\right) P_r\left(\frac{2}{L_z}z\right). \tag{12}$$

Here $H(\mathbf{r})=1$ in all ion-accessible regions and $H(\mathbf{r})=0$ otherwise. L_x, L_y, and L_z specify the length of the LGCMC region in each direction and n indicates the nth index in the $(0,0,0),(0,0,1),\cdots,(p,q,r),\cdots,(N_p,N_q,N_r)$ indices, where N_q, N_q, and N_r are the upper limits of the Legendre polynomials. Then, the basis-set potentials $\phi_n^{\mathrm{rf}}(\mathbf{r})$ corresponding to the basis functions $b_n(\mathbf{r})$ in Eq. (8) can be written as

$$\phi_n^{\mathrm{rf}}(\mathbf{r}) = \phi_n^{\mathrm{inh}}(\mathbf{r}) - \phi_n^{\mathrm{blk}}(\mathbf{r}). \tag{13}$$

In Eq. (13), $\phi_n^{\mathrm{inh}}(\mathbf{r})$ represents the electrostatic potential due to the basis function $b_n(\mathbf{r})$ in the system with inhomogeneous dielectric boundaries and can be calculated using

$$\nabla \cdot (\epsilon(\mathbf{r}) \nabla \phi_n^{\mathrm{inh}}(\mathbf{r})) - \epsilon_w \kappa_{\mathrm{I(II)}}^2(\mathbf{r}) \phi_n^{\mathrm{inh}}(\mathbf{r}) = -b_n(\mathbf{r}), \tag{14}$$

where the screening factor $\kappa_{\mathrm{I(II)}}$ is set to zero in the LGCMC region where ions are explicitly simulated.[9] The electrostatic potential, $\phi_n^{\mathrm{blk}}(\mathbf{r})$ is expressed in terms of basis functions in the homogeneous bulk water system by

$$\epsilon_w \nabla^2 \phi_n^{\mathrm{blk}}(\mathbf{r}) = -b_n(\mathbf{r}). \tag{15}$$

For more details, See Ref. 9.

C. Definition of the time step and calculation of the ion current

The time step is required to calculate ion currents in the KLGCMC/MF algorithm. This can be evaluated from the well-known Einstein-Smoluchowski equation,[36]

$$\Delta t = \frac{(\Delta L)^2}{6D}, \tag{16}$$

where ΔL is the lattice spacing in the KLGCMC/MF simulation and D is an ion diffusion coefficient. In the present KLGCMC/MF simulation, we use the same diffusion coefficient for K^+ and Cl^- ions. (See Table I). Then, ion currents are calculated by counting the ions crossing a cross section of the ion channel per unit time step and by averaging the number of counted ions,

$$I_{\mathrm{ion}} = \frac{\langle N_{\mathrm{ion}}^{\mathrm{cross}} \rangle}{\Delta t} = \frac{6D\langle N_{\mathrm{ion}}^{\mathrm{cross}} \rangle}{(\Delta L)^2}, \tag{17}$$

where $N_{\mathrm{ion}}^{\mathrm{cross}}$ is the number of ions (cations or anions) crossing the cross section per time step.

In the current KLGCMC/MF simulation, one KLGCMC cycle, equivalent to one time step, is composed of 20 creation or deletion attempts in the LC regions and move attempts for all the ions in the LGCMC region.

III. DATA AND RESULTS

A. Description of the model system and details of the KLGCMC simulations

A two-dimensional cross section of the 3D model ion channel is presented in Fig. 1. To model a cation-selective ion channel, we place 64 dipoles inside the channel (8 dipoles around the channel axis and 8 rings along the channel axis) with the positive ends closer to the channel/water interface. The length of one dipole is 5 Å and the charges of the positive and negative ends are $\pm 0.08e$, respectively.

We take the water dielectric constant to be $80\epsilon_0$ both in bulk and inside the channel. This assumes that the dielectric constant of water inside the channel is the same as in the bulk, which is an assumption which needs to be tested in general. However, the purpose of this study is to investigate the reaction field effect due to the lipid bilayer using the KLGCMC simulation method, so we have not explored the effect of varying the water dielectric constant. The channel and membrane are assumed to have the same dielectric constant ϵ_m, which is taken to be $\epsilon_m=80\epsilon_0$ or $\epsilon_m=2\epsilon_0$ to investigate the reaction field effect. The diffusion coefficients of both K^+ and Cl^- are taken to be 2.0×10^{-5} cm^2/s in bulk and

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inside the channel. We use a Stern layer parameter of 1.6 Å that is close to the radius of the K^+ ion because many more K^+ ions than Cl^- are found inside the channel, and the interaction of K^+ ions with the channel is more important than with Cl^- ions. The LJ parameters and other parameters used in this study are presented in Table I. The parameter B in Eq. (2), associated with the bulk ion concentration, is determined by performing LGCMC simulations for a bulk ion system with different B values and by selecting values that give a bulk ion concentration of $0.1M$ or $0.5M$.[26]

The electrostatic focusing method is used to solve the modified PB equations for the electrostatic potential due to the interaction between the ions and the fixed charges in the channel and membrane in Eqs. (7a)–(7c) and the basis-set electrostatic potentials in Eq. (13). In that method, the electrostatic potentials for the whole system including the MF and LGCMC regions are calculated on a $71\times71\times101$ grid with a grid size of 1.0 Å, followed by a finer grid of a $81\times81\times141$ with a grid size of 0.5 Å only in the LGCMC region. In general, a total of 225 basis functions from the combination of five, five, and nine Legendre polynomials in x, y, and z directions, respectively, are used for calculation of the basis-set electrostatic potentials. An optimal cutoff eigenvalue $s_{\min}$ to eliminate linear dependency of the eigenvalues of $\mathbf{S}$ in calculating the inverse of the matrix $\mathbf{S}$ in Eq. (9) is taken to be 0.001.[9]

The KLGCMC/MF simulations with explicit ions are performed on a $81\times81\times141$ grid with a grid size of 0.5 Å. Reflecting boundary conditions are used at the boundaries between the LGCMC region and the MF regions in the x, y, and z directions. If the size of LC region is too small, the correct number of ions are not maintained and if the size of LC region is large enough to include the channel region, the fluctuation of ion currents is huge due to the creation or deletion of ions inside the channel. The size of each LC region is 5 Å in this study.

For calculating the ion current at each applied voltage, the system reaches a steady-state after 5 000 000 KLGCMC cycles, and the data for the ion current calculation are collected for another 65 000 000 KLGCMC cycles.

B. Test of the basis-set expansion method for the reaction field in the model system with $\epsilon_m=2\epsilon_0$

To investigate the treatment of the reaction field by the basis-set expansion method, we first turn off all the charges of the dipoles in the ion channel and generate a configuration by randomly distributing cations and anions in the LGCMC region. The number of cations and anions varies around 8, but the total number of ions is maintained at 16. We place more than two cations inside the model ion channel in each configuration with the assumption that the ion channel is cation selective. Overall, 50 random configurations are generated in this way. The reaction field energy of the system in each configuration can be calculated by

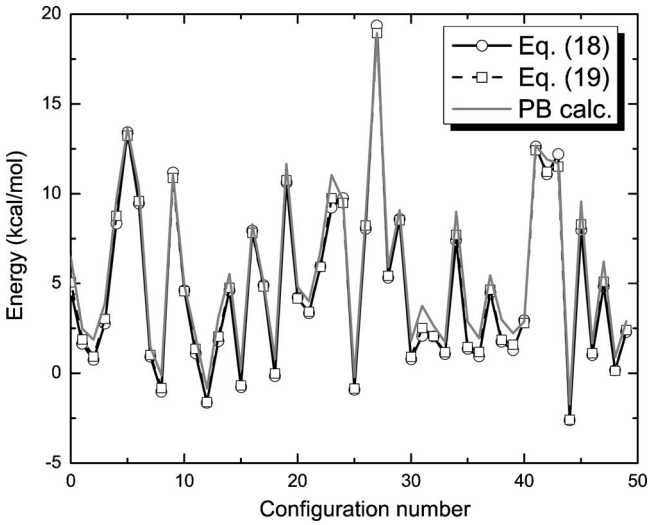


FIG. 2. Reaction field energy calculations for 50 independent configurations taken from the basis-set expansion results and from the PB equation results for the model system. For the basis-set expansion method, both Eqs. (18) and (19) are used.

$$U_{\mathrm{rf}}=\tfrac{1}{2}\sum_i q_i\phi_{\mathrm{rf}}(\mathbf{r}_i), \tag{18}$$

where $\phi_{\mathrm{rf}}(\mathbf{r}_i)$ is the reaction field induced by ion i at position $\mathbf{r}_i$ that is given by Eq. (8). Another expression for the reaction field energy can be written as

$$U_{\mathrm{rf}}=\tfrac{1}{2}\mathbf{Q}^T\mathbf{M}^*\mathbf{Q}, \tag{19}$$

where the vector $\mathbf{Q}$ is given by Eq. (11) and the matrix $\mathbf{M}^*$ is given as

$$\mathbf{M}^*=(\mathbf{S}^{-1})^T\mathbf{M}\mathbf{S}^{-1}. \tag{20}$$

Here the matrix $\mathbf{S}$ is given by Eq. (10), $(\mathbf{S}^{-1})^T$ is the transpose matrix of $\mathbf{S}^{-1}$, and the matrix $\mathbf{M}$ is calculated by

$$M_{mn}=\int d\mathbf{r}d\mathbf{r}'b_m(\mathbf{r})G_{\mathrm{rf}}(\mathbf{r},\mathbf{r}')b_n(\mathbf{r}')=\int d\mathbf{r}b_m(\mathbf{r})\phi_n^{\mathrm{rf}}(\mathbf{r}), \tag{21}$$

where $\phi_n^{\mathrm{rf}}(\mathbf{r})$ is given by Eq. (13).

The reaction field energies of the 50 configurations are calculated using both Eqs. (18) and (19) and compared with those obtained from the finite difference PB equation. Figure 2 shows good agreement between the basis-set expansion method and the PB calculation. It also shows that the energies obtained from Eqs. (18) and (19) are nearly indistinguishable. Assuming that the PB calculations are exact, the average errors over the 50 configurations from Eqs. (18) and (19) are 0.795 and 0.707 kcal/mol, respectively. Note that, in all 50 configurations, more than two cations are always placed inside the channel. In the KLGCMC/MF simulations shown later, however, less than one cation is generally found inside the channel. Therefore, the error in the simulations should be smaller than that reported above. In configuration 27, there are three cations found inside the channel, and the reaction field energy of that configuration is largest among the 50 configurations.

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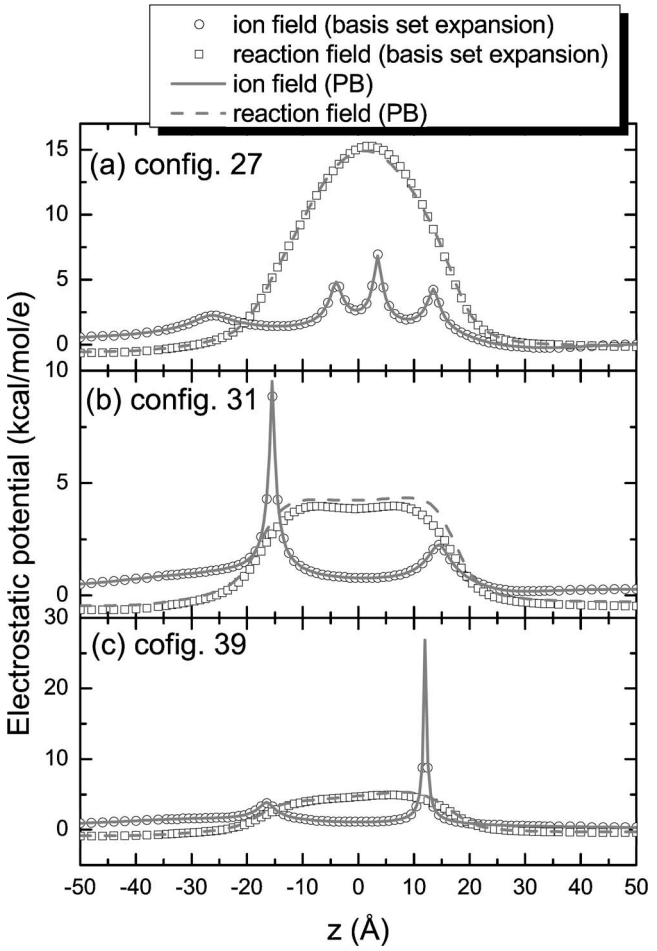


FIG. 3. Electrostatic potentials of the ions (ion field) and induced charges (reaction field) from the basis-set expansion method for three configurations: (a) configuration 27, (b) configuration 31, and (c) configuration 39. Comparison with the PB calculations is also shown.

For configurations 27, 31, and 39, the electrostatic potentials due to the ions [Eq. (6)] and the induced charges [Eq. (8)] on the channel/water and membrane/water boundaries are shown in Fig. 3. The electrostatic potentials for the basis-set expansion method are calculated using Eq. (8). In configurations 31 and 39, the reaction field potentials for the basis-set expansion method show small deviations from the PB results, and these deviations lead to the errors in the calculation of the reaction field energy with the basis-set expansion method.

C. Ion current calculations for the model system where the bulk concentrations at the bottom and top are both 0.1*M*

In this example, the bulk concentrations at the bottom and top are both $0.1M$. To investigate the reaction field effect on ion currents, both $80\epsilon_0$ and $2\epsilon_0$ are used as the dielectric constant of the membrane and channel (ϵ_m).

Figure 4 shows the ion current-voltage curves in the case of $\epsilon_m=80\epsilon_0$. Because the dielectric constant of the membrane and channel is the same as water, no charges are induced on the channel/water or the membrane/water boundaries, and no reaction field effect appears. Due to the absence of the reaction field, good agreement between the KLGCMC/MF results and the PNP calculations is found; the K^+ and Cl^- conductances from the KLGCMC/MF simulations are 73.70 and 22.16 pS, respectively, and those from the PNP calculations are 77.03 and 22.80 pS, respectively.

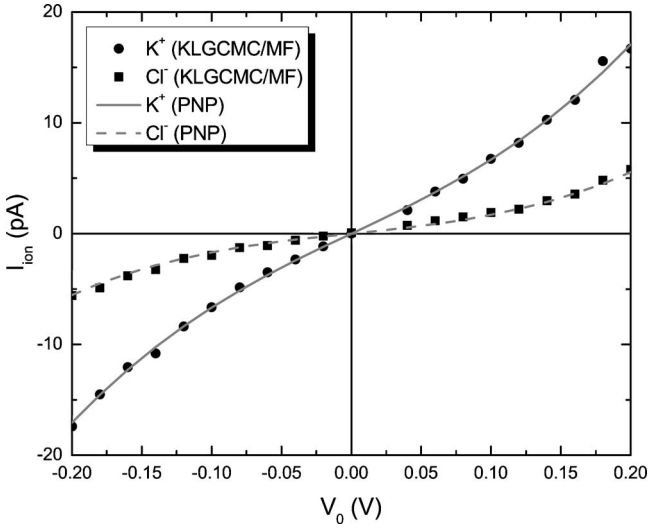


FIG. 4. Ion current-voltage curves of the K^+ and Cl^- ions from the KLGCMC/MF simulations in the model ion channel system in the case of $\epsilon_m=80\epsilon_0$. The bulk ion concentrations at the bottom and top are both $0.1M$. Comparison with the PNP calculations is also shown.

The electrostatic potentials of the ions (ion field), fixed dipoles on the channel (static field), and induced charges (reaction field) at $V_0=-0.2$ V are presented in Fig. 5(a). The total electrostatic potential from the KLGCMC/MF simula-

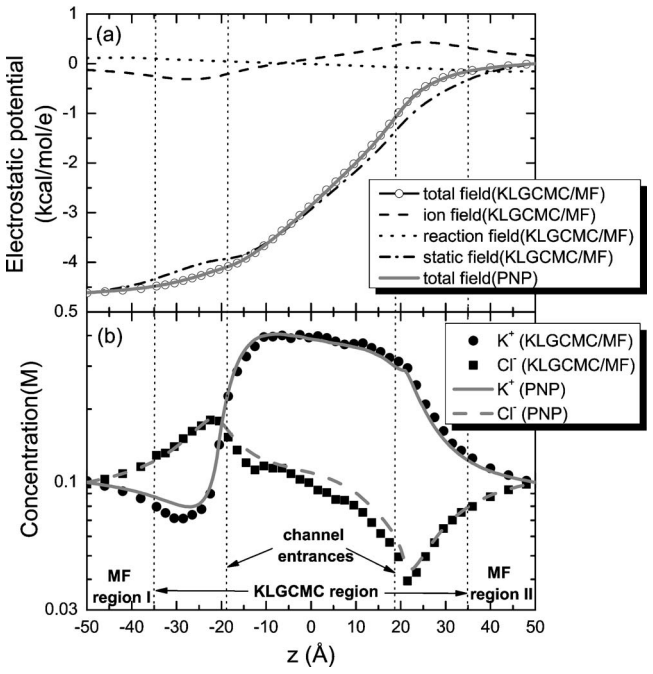


FIG. 5. (a) Electrostatic potentials due to the total, ions (ion field), induced charges (reaction field), and fixed dipoles on the channel (static field) and (b) K^+ and Cl^- concentrations at $V_0=-0.2$ V. The membrane and channel dielectric constant and bulk ion concentrations are the same as in the previous figure. The dotted lines represent the boundaries between the LGCMC and MF regions at $z=\pm35.0$ Å and the channel entrances at $z=\pm19.0$ Å. Comparison with the PNP calculations is also shown. Note the semilogarithmic scale on the y axis in (b).

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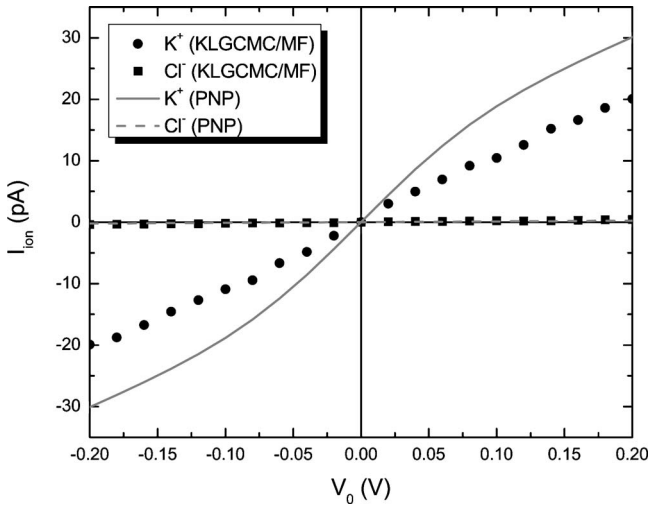


FIG. 6. Ion current-voltage curves of the K^+ and Cl^- ions from the KLGCMC/MF simulations in the model ion channel system in the case of $\epsilon_m=2\epsilon_0$. The bulk ion concentrations at the bottom and top are both $0.1M$. Comparison with the PNP results is also shown.

tion shows excellent agreement with that of the PNP calculation. At $V_0=-0.2$ V, K^+ ions flow from the top to bottom. Because of the small size of the model ion channel, however, the flow of K^+ ions is partially blocked, and K^+ ions accumulate at the top side. For the same reason, Cl^- ions are accumulated at the bottom. As a result, the electrostatic potential from the ion field is positive at the top and negative at the bottom. There is no electrostatic potential due to the reaction field of the induced charges on the channel/water or membrane/water bound aries because the channel and membrane dielectric constant ($\epsilon_m=80\epsilon_0$) is the same as the water dielectric constant ($\epsilon_w=80\epsilon_0$) in this example. However, there is a small amount of the electrostatic potential due to the reaction field at the boundaries between the LGCMC region and the MF regions. The negative electrostatic potential of the reaction field at the top occurs because K^+ ions accumulated at the top induce counterions in MF region II where ions are represented by $\kappa_{\mathrm{II}}(\mathbf{r})$. The negative electrostatic potential of the reaction field due to counterions in the MF region II also stabilizes the cations close to the boundary and minimizes the surface effect which in general reduces the number of ions at the surface or boundary due to insufficient solvation by counterions. For the same reason, Cl^- ions at the bottom induce counterions in the MF region I and the reaction field by the induced counterions leads to the positive electrostatic potential at the bottom. Figure 5(b) shows the K^+ and Cl^- concentration profiles from the KLGCMC/MF simulation at $V_0=-0.2$ V. This shows good agreement with the PNP calculations. As mentioned above, K^+ ions are accumulated at the top and Cl^- ions are accumulated at the bottom. Because the model ion channel is cation selective, more K^+ ions are found inside the channel. The average numbers of K^+ and Cl^- obtained from the KLGCMC/MF simulation are 0.142 and 0.0380, respectively; those numbers from the PNP calculation are 0.142 and 0.0421, respectively.

Ion current-voltage curves for $\epsilon_m=2\epsilon_0$ are presented in Fig. 6. There is considerable difference in this case between

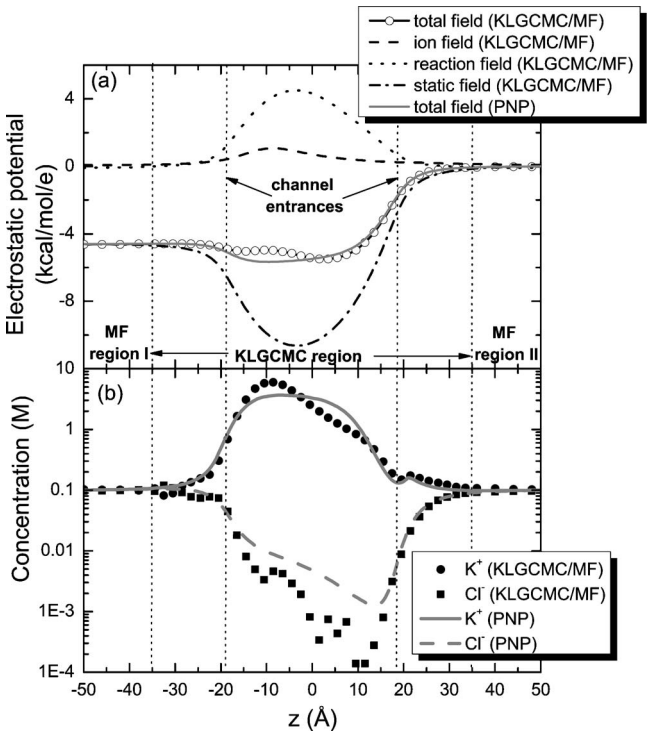


FIG. 7. (a) Electrostatic potentials due to the total, ions (ion field), induced charges (reaction field), and fixed dipoles on the channel (static field) and (b) K^+ and Cl^- concentrations at $V_0=-0.2$ V. The membrane and channel dielectric constant and bulk ion concentrations are the same as in the previous figure. The dotted lines represent the boundaries between the LGCMC and MF regions at $z=\pm35.0$ Å and the channel entrances at $z=\pm19.0$ Å. Comparison with the PNP calculations is also shown. Note the semilogarithmic scale on the y axis in (b).

the KLGCMC/MF simulation results and the PNP calculations. Owing to the much lower dielectric constant of the channel and membrane ($\epsilon_m=2\epsilon_0$) than the water dielectric constant ($\epsilon_w=80\epsilon_0$) in this case, an ion inside the channel or entering the channel induces charges on the channel/water or membrane/water boundaries with the same sign as the ion charge. This leads to repulsive interactions between the ion and the induced charges, resulting in a reduction of the ion currents in the channel. The KLGCMC/MF simulation fully accounts for the reaction field effect, whereas PNP theory, which lacks the exact description of the reaction field, overestimates the ion currents. The conductance of the K^+ ions from the KLGCMC/MF simulations is 105.38 pS, whereas that from the PNP calculation is 165.55 pS.

In Fig. 7(a), the electrostatic potentials of various sources at $V_0=-0.2$ V are shown. The reaction field electrostatic potential due to the induced charges is highly positive because only K^+ ions are selectively allowed to enter the channel and the K^+ ions induce positive charges on the channel/water boundary. The static field electrostatic potential from the fixed dipoles in the channel has a deeper energy well compared to that in the previous example because the lower dielectric constant of the channel and membrane causes a stronger static field. A much higher concentration of K^+ ions than Cl^- at $V_0=-0.2$ V is found inside the channel in Fig. 7(b) because of the cation-selective characteristic of the ion channel. A stronger static field than the previous example leads to a higher concentration of K^+ ions. The average num-

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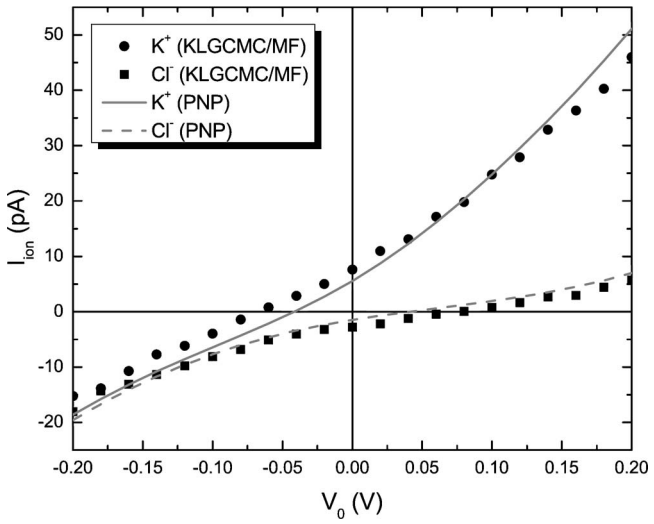


FIG. 8. Ion current-voltage curves for the K^+ and Cl^- ions from the KLGCMC/MF simulations in the model ion channel system in the case of $\epsilon_m=80\epsilon_0$. The bulk ion concentrations at the bottom and top are $0.5M$ and $0.1M$, respectively. Comparison with the PNP results is also shown.

bers of K^+ and Cl^- from the KLGCMC/MF simulation are 0.949 and 0.001 69, respectively, while the numbers from the PNP calculation are 0.870 and 0.002 96, respectively. The KLGCMC/MF simulation predicts a higher concentration of K^+ ions and a lower concentration of Cl^- ions inside the channel than PNP calculation does.

D. Ion current calculations for the model system where the bulk concentrations at the bottom and top are 0.5*M* and 0.1*M*, respectively

In this example, we use asymmetric bulk concentrations at the bottom and top: $0.5M$ at the bottom and $0.1M$ at the top. To investigate the reaction field effect on the ion current, we also use two different channel and membrane dielectric constants (ϵ_m): $80\epsilon_0$ and $2\epsilon_0$.

Figure 8 presents ion current-voltage curves from the KLGCMC/MF simulations and PNP calculations for $\epsilon_m=80\epsilon_0$. The KLGCMC simulations are in good agreement with the PNP calculations except that the cationic currents from the simulation are slightly lower than those from the PNP calculations. The K^+ and Cl^- conductances from the KLGCMC/MF simulations are 141.56 and 46.91 pS, respectively; those from the PNP calculations are 166.01 and 58.21 pS, respectively. The small current at $V_0=0$ V is due to the ion concentration gradient between the bottom and top. In Fig. 9(a), comparison of the electrostatic potentials between the KLGCMC simulation and the PNP calculation at $V_0=-0.2$ V is also found to show good agreement. However, there is a discrepancy between the KLGCMC/MF simulation results and the PNP calculations in the ion concentration profiles plotted in Fig. 9(b). The average numbers of K^+ and Cl^- ions inside the channel predicted by the KLGCMC/MF simulation is 0.154 and 0.117, respectively; those from the PNP calculations are 0.172 and 0.140, respectively.

Figure 10 shows the K^+ and Cl^- ion currents for $\epsilon_m=2\epsilon_0$. Compared with the ion currents with $\epsilon_m=80\epsilon_0$ in the previous example, the KLGCMC/MF simulation predicts a slight increase in the K^+ current and a negligible value of the Cl^- current. The PNP calculation also predicts minimal Cl^- currents, but it overestimates the K^+ currents in comparison with the KLGCMC/MF simulation. The KLGCMC/MF simulation presents a current-voltage relationship that is closer to being linear than the PNP calculation. The K^+ and

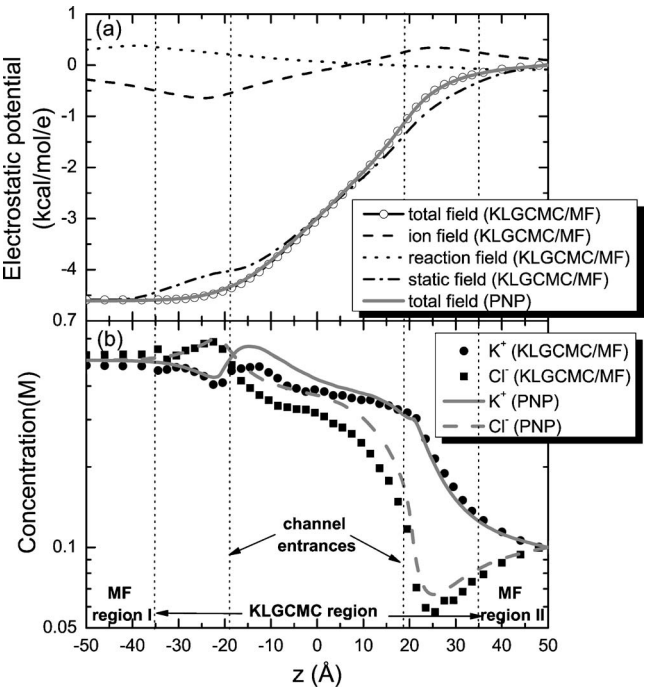


FIG. 9. (a) Electrostatic potentials due to the total, ions (ion field), induced charges (reaction field), and fixed dipoles on the channel (static field) and (b) K^+ and Cl^- concentrations at $V_0=-0.2$ V. The membrane and channel dielectric constant and bulk ion concentrations are the same as in the previous figure. The dotted lines represent the boundaries between the LGCMC and MF regions at $z=\pm35.0$ Å and the channel entrances at $z=\pm19.0$ Å. Comparison with the PNP calculations is also shown. Note the semilogarithmic scale on the y axis in (b).

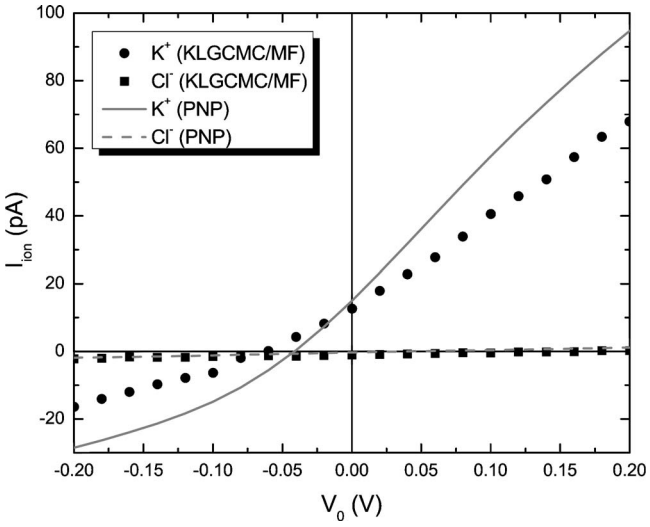


FIG. 10. Ion current-voltage curves of the K^+ and Cl^- ions from the KLGCMC/MF simulations in the model ion channel system in the case of $\epsilon_m=2\epsilon_0$. The bulk ion concentrations at the bottom and top are $0.5M$ and $0.1M$, respectively. Comparison with the PNP calculations is also shown.

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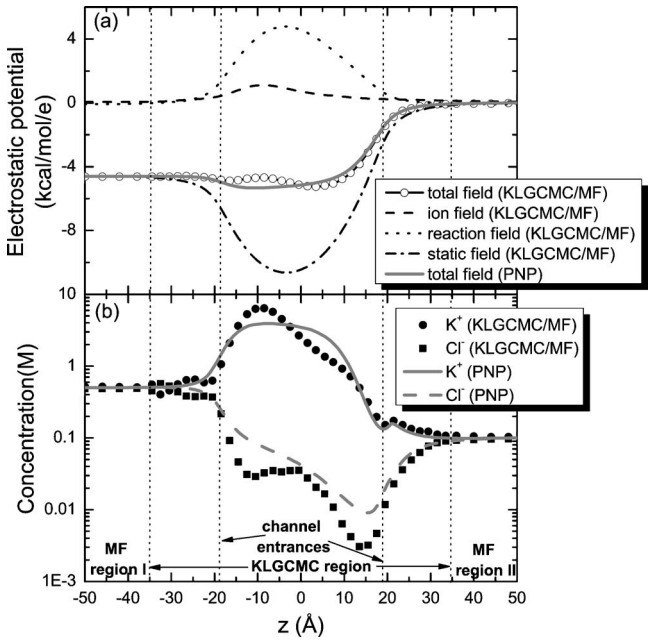


FIG. 11. (a) Electrostatic potentials due to the total, ions (ion field), induced charges (reaction field), and fixed dipoles on the channel (static field) and (b) K^+ and Cl^- concentrations at $V_0=-0.2$ V. The membrane and channel dielectric constant and bulk ion concentrations are the same as in the previous figure. The dotted lines represent the boundaries between the LGCMC and MF regions at $z=\pm 35.0$ Å and the channel entrances at $z=\pm 19.0$ Å. Comparison with the PNP calculations is also shown. Note the semilogarithmic scale on the y axis in (b).

Cl^- conductances obtained from the KLGCMC/MF simulations are 210.43 and 5.95 pS, while those from the PNP calculations are 330.27 and 7.71 pS, respectively. The electrostatic potentials from various sources at $V_0=-0.2$ V are presented in Fig. 11(a). A strong reaction field is produced due to the induced charges on the channel/water boundary by ions inside the channel. Much higher concentrations of K^+ ions than Cl^- ions are shown in Fig. 11(b). The average numbers of K^+ and Cl^- ions from the KLGCMC/MF simulation are 1.03 and 0.012, respectively, whereas those from the PNP calculation are 0.97 and 0.02, respectively.

IV. DISCUSSION AND CONCLUSION

In this paper, we have presented a new lattice-based MC simulation method, KLGCMC/MF, which makes it possible to describe reaction field effects in ion channel transport, and we have used this method to calculate ion currents in a model ion channel system. In the KLGCMC/MF method, the ion channel part of the system is treated with a KLGCMC simulation, while the rest is described by PB mean field theory. Based on the on-lattice approach, the KLGCMC/MF simulation allows one to describe long-time dynamics by avoiding any interpolation schemes. The reaction field potential that is produced by induced charges on the channel/water and membrane/water boundary was calculated using a basis-set expansion method. PB and PNP calculations, both based on dielectric continuum models, were also performed to test the basis-set expansion method and compare with the KLGCMC/MF simulation results.

Comparison between the basis-set expansion and direct PB theory results for the calculation of the reaction field shows that the basis-set expansion method describes the reaction field energies and potentials very well, even when more than two ions are found inside the ion channel.

In the case where the channel and membrane have the same dielectric constant as water ($\epsilon_m=80\epsilon_0$), the reaction field effect is minimal, and the KLGCMC/MF simulation results are in excellent agreement with the PNP calculations. When the dielectric constant of the channel and membrane is much lower than water ($\epsilon_m=2\epsilon_0$), however, the KLGCMC/MF simulation predicts smaller ion currents than the PNP calculations. This indicates that PNP theory does not describe the reaction field properly and overestimates the ion currents. Graf *et al.* proposed a modified and improved PNP method called dielectric self-energy PNP (DSEPNP) method to include the reaction field effect in the original PNP theory.[14] It will be useful to compare the current KLGCMC simulation results with the DSEPNP calculations in future work.

In the case where the dielectric constants of the channel and membrane are the same as water ($\epsilon_m=80\epsilon_0$), there is a slight reaction field at the boundaries between the KLGCMC region and the MF regions. That electrostatic potential results from counterions induced in the MF regions by the explicit ions in the KLGCMC region. When the channel and membrane dielectric constant is lower than the water constant ($\epsilon_m=2\epsilon_0$), the electrostatic potential due to induced charges is much more important than the electrostatic potential due to the ions.

Ion selectivity is found in the ion concentration profiles: high concentrations of K^+ ions and low concentrations of Cl^- ions. In the case where the dielectric constant of the channel and membrane is lower than water ($\epsilon_m=2\epsilon_0$), the KLGCMC/MF simulation predicts a higher concentration of K^+ ions and a lower concentration of Cl^- ions inside the channel than the PNP calculation does. In the KLGCMC/MF simulations, the average number of cations inside the ion channel is mostly less than 1, which implies that ion correlation effects on the currents are small.

This KLGCMC/MF simulation study clearly shows how ions are selectively translocated through an ion channel in the cell. Despite being designed to be cation selective, an ion channel and membrane with the same dielectric constant as water allows a considerable amount of Cl^- to go through the channel. If the channel and membrane dielectric constant is much lower than water, however, the channel lets only cations pass through the channel, and minimal anion currents are observed even at high ion concentrations. This demonstrates two important factors in selective ion transport in the ion channel: low dielectric media such as protein ion channels and membranes and the fixed charge alignment on the channel.

In this study, variation of the ion diffusion coefficients inside the channel was not taken into account. Also ignored are fluctuations in the structure of the ion channel. Although these issues should be addressed in higher level simulations, we believe that the KLGCMC/MF simulation method presented in this paper elucidates many important and interest-

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ing issues raised in the study of ion transport in ion channels. An application of the KLGCMC/MF simulation method to realistic ion channels is under development.

ACKNOWLEDGMENTS

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[1] B. Hille, *Ionic Channels of Excitable Membranes* (Sinauer Associates Inc., Sunderland, MA, 1991).
[2] F. M. Ashcroft, Nature (London) **440**, 440 (2006).
[3] V. A. Parsegian, Nature (London) **221**, 844 (1969).
[4] D. Chen, J. D. Lear, and R. S. Eisenberg, Biophys. J. **72**, 97 (1997).
[5] B. Roux and R. MacKinnon, Science **285**, 100 (1999).
[6] B. Corry, S. Kuyucak, and S. H. Chung, Biophys. J. **78**, 2364 (2000).
[7] M. G. Kurnikova, R. D. Coalson, P. Graf, and A. Nitzan, Biophys. J. **76**, 642 (1999).
[8] P. Graf, A. Nitzan, M. G. Kurnikova, and R. D. Coalson, J. Phys. Chem. B **104**, 12324 (2000).
[9] W. Im and B. Roux, J. Chem. Phys. **115**, 4850 (2001).
[10] W. Im and B. Roux, J. Mol. Biol. **322**, 851 (2002).
[11] T. A. van der Straaten, J. Tang, U. Ravaioli, R. S. Eisenberg, and N. R. Aluru, J. Comput. Electron. **2**, 29 (2003).
[12] T. van der Straaten, G. Kathawala, and U. Ravaioli, J. Comput. Electron. **2**, 231 (2003).
[13] R. S. Eisenberg, Biophys. Chem. **100**, 507 (2003).
[14] P. Graf, M. G. Kurnikova, R. D. Coalson, and A. Nitzan, J. Phys. Chem. B **108**, 2006 (2004).
[15] M. H. Cheng, M. Cascio, and R. D. Coalson, Biophys. J. **89**, 1669 (2005).
[16] M. H. Cheng and R. D. Coalson, J. Phys. Chem. B **109**, 488 (2005).
[17] H. Hwang, G. C. Schatz, and M. A. Ratner, J. Phys. Chem. B **110**, 6999 (2006).
[18] S. Edwards, B. Corry, S. Kuyucak, and S.-H. Chung, Biophys. J. **83**, 1348 (2002).
[19] B. Corry, S. Kuyucak, and S. H. Chung, Biophys. J. **84**, 3594 (2003).
[20] A. B. Mamonov, R. D. Coalson, A. Nitzan, and M. Kurnikova, Biophys. J. **84**, 3646 (2003).
[21] G. R. Smith and S. P. Sansom, Biophys. J. **75**, 2767 (1998).
[22] T. W. Allen, S. Kuyucak, and S.-H. Chung, Biophys. J. **77**, 2502 (1999).
[23] T. W. Allen, O. S. Andersen, and B. Roux, Proc. Natl. Acad. Sci. U.S.A. **101**, 117 (2004).
[24] P. S. Crozier, R. L. Rowley, N. B. Holladay, D. Henderson, and D. D. Busath, Phys. Rev. Lett. **86**, 2467 (2001).
[25] G. Moy, B. Corry, S. Kuyucak, and S. H. Chung, Biophys. J. **78**, 2349 (2000).
[26] W. Im, S. Seefeld, and B. Roux, Biophys. J. **79**, 788 (2000).
[27] M. Hoyles, S. Kuyucak, and S.-H. Chung, Phys. Rev. E **58**, 3654 (1998).
[28] D. Boda, D. Gillespie, W. Nonner, D. Henderson, and B. Eisenberg, Phys. Rev. E **69**, 046702 (2004).
[29] J. P. Valleau and L. K. Cohen, J. Chem. Phys. **72**, 5935 (1980).
[30] G. M. Torrie and J. P. Valleau, J. Chem. Phys. **73**, 5807 (1980).
[31] G. M. Torrie and J. P. Valleau, J. Phys. Chem. **86**, 3251 (1982).
[32] A. Papadopoulou, E. D. Becker, M. Lupkowski, and F. van Swol, J. Chem. Phys. **98**, 4897 (1993).
[33] B. Roux, Biophys. J. **73**, 2980 (1997).
[34] B. Roux, Biophys. J. **77**, 139 (1999).
[35] W. Im, S. Berněche, and B. Roux, J. Chem. Phys. **114**, 2924 (2001).
[36] P. W. Atkins, *Physical Chemistry*, 4th ed. (Oxford University Press, Oxford, 1990).
[37] B. Roux, Biophys. J. **71**, 3177 (1996).

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